Today:

(semigroup property of $n \mapsto p_n$)

- multistep trans. prob. & Chapman-Kolmogorov equation

$$p_n(x,y) := P(X_n = y \mid X_0 = x)$$

- stopping times, - hitting time of a state
  - return " to " "

- strong Markov property: Markov property holds at a stopping time.

- transience/recurrence of a state:

"will I ever see you again?"

classification of states in finite state MC.

- stationary distributions: how much of your time is spent at x, in the long run.

Chapman-Kolmogorov

Let $p_n(x,y) = P(X_n = y \mid X_0 = x)$ n-step trans. prob.

We want to show

$$\forall n, \quad p_n = (p_1)^n \quad \text{(as matrices).}$$

Enough to show $\forall n, m$,

$$p_{n+m} = p_n p_m. \quad \text{(semigroup property)}$$

for any x, y, n, m,

$$p_{n+m}(x,y) = P(X_{n+m} = y \mid X_0 = x)$$

$$= \sum_{z \in S} P(X_{n+m} = y \mid X_n = z, X_0 = x) P(X_n = z \mid X_0 = x)$$

uses $P(B \cap A) = P(B \mid A) P(A)$

& $P(A) = \sum_{B \in \mathcal{P}} P(B \cap A)$

$\mathcal{P}$ = partition of Ω

$$= \sum_{z \in S} \underbrace{P(X_{n+m} = y \mid X_n = z)}_{= p_m(z,y) \text{ for (time-homogeneous) MC.}} P(X_n = z \mid X_0 = x)$$

$$= \sum_{z \in S} p_m(z,y)\, p_n(x,z)$$

$$= \sum_{z \in S} p_n(x,z)\, p_m(z,y)$$

$$= \underbrace{(p_n p_m)}_{\text{product of matrices } p_n \,\&\, p_m}{}_{\underbrace{xy}_{(x,y)\text{ entry}}}$$

Hitting & return time

text denotes T_x^+ as T_x in §1.3.

Let $T_x = \min\{n \geq 0 : X_n = x\}$ hitting time of x

$T_x^+ = \min\{n > 0 : X_n = x\}$ return time to x.

T_x may $= 0$, but

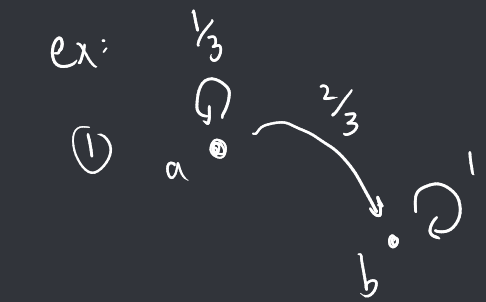
$T_x^+ \geq 1$.

using $\min \emptyset := \infty$

define $\rho_{xx} = P(T_x^+ < \infty \mid X_0 = x)$

return probability.

ex:



find ρ_{aa} for each.

① $p_{aa} = P(X_1 = a \mid X_0 = a) = p(a,a) = 1/3.$

$$\left(\text{since if } X_1 = \begin{cases} a, & T_x^+ = 1 \\ b, & T_x^+ = \infty \end{cases} \right)$$

② $p_{aa} = 1$ since if $X_0 = a$, then $X_2 = a$

w.p. 1

"with probability 1"

a.k.a. "almost surely."

Say that x is recurrent if $\rho_{xx} = 1$

transient if $\rho_{xx} < 1$.

<u>Claim</u>: for a MC, if x: $\rho_{xx} < 1$

then the random variable

$$N := \#\{n \geq 0 : X_n = x\}$$

has dist$^{\underline{bn}}$ $\underbrace{\text{geometric}(1-\rho_{xx})}$

if $X_0 = x$.

trials to first "success" if "success" has probab. $1-\rho_{xx}$.

Explanation:

if $X_0 = x$ ---- return, or never return.

if never return, no more visits.

if return to x, "start fresh".

strong Markov property.

in discrete time, Markov property $\Rightarrow$ strong Markov property

What is strong Markov?

notice that for any n, $\{T_x^+ \leq n\}$ can be determined from $X_0, \dots, X_n$.

i.e. whether or not returned by time n does not depend on future values of MC.

this property defines a "stopping time".

Strong Markov property: if (X_n) is a MC &

T is a stopping time then $\forall\ x_0, \dots, x_k$,

$$P(X_{n+i} = x_{n+i}\ \forall\ i \in \{1, \dots, k\} \mid T = n,\ X_n = x_0)$$

$$= P(X_i = x_i\ \forall\ i \in \{1, \dots, k\} \mid X_0 = x_0)$$

$$= \prod_{i=1}^{k} p(x_{i-1}, x_i)$$

wts strong Markov gives geometric distbtn. for # visits to transient state.

Let $T^+_{x,n}$ = time of nth return to x

(exercise: $= \min\{ \cdots \}$).

(*) $P(T^+_{x,n+1} < \infty \mid T_{x,n} < \infty)$

implies $X_m = x$

$$= \sum_m \underbrace{P(T^+_{x,n+1} < \infty \mid T_{x,n} = m)}_{\text{by Strong Markov property}} P(T_{x,n} = m \mid T_{x,n} < \infty)$$

$$= P(T_x^+ < \infty \mid X_0 = x)$$

$$= \sum_m P(T_x^+ < \infty \mid X_0 = x)\, \underbrace{P(T_{x,n} = m \mid T_{x,n} < \infty)}_{\text{add up to } 1}$$

$$= P(T_x^+ < \infty \mid X_0 = x) \cdot 1.$$

visits has $N > n$ iff $T_{x,n}^+ < \infty$.

From (*), this has prob. $= \Big(\underbrace{P(T_x^+ < \infty \mid X_0 = x)}_{\rho_{xx}}\Big)^n$

i.e. $P(N > n) = \rho_{xx}^{n} \quad \forall n$

which characterizes $\mathrm{geom}(1-\rho_{xx})$.

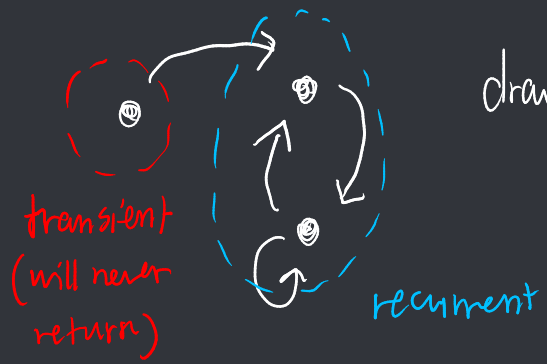
Stronger property: "excursions", indexed by n, as

$$\left(X_{T_n + i} \right)_{i=0}^{T_{n+1} - T_n - 1}$$

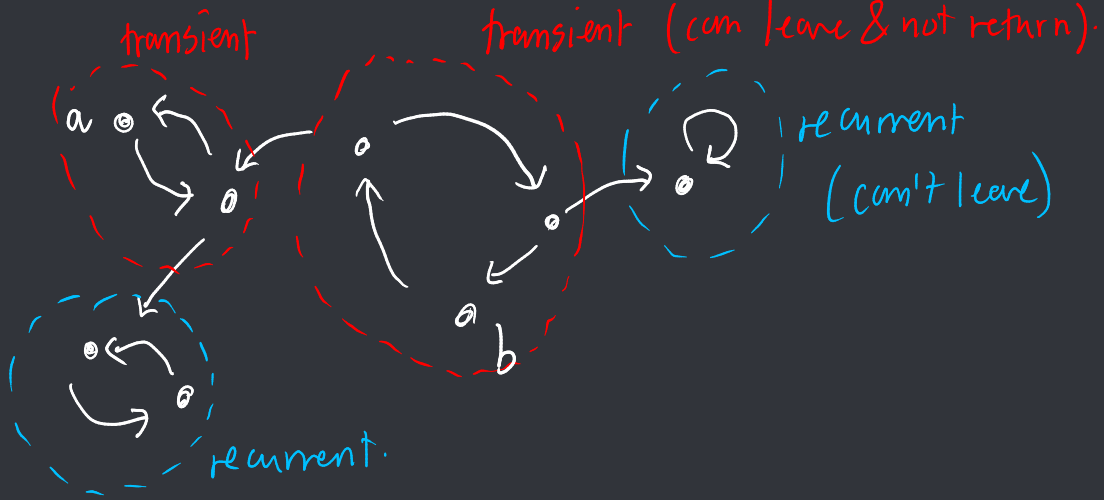
are i.i.d. random "vectors"
(w/ random # entries).

Which states in a MC are transient & which are recurrent?

Reading it from the directed graph of trans. matrix.



draw arrows (x,y) for which $p(x,y) > 0$.



Formally: say that $x \longrightarrow y$ (y accessible from x)

if $\exists\, n > 0 : p_n(x, y) > 0$

and $x \not\to y$ if not.

In above, does $a \to b$? No, $a \not\to b$.

does $b \to a$? Yes, $b \to a$.

Picture $\to$ statement

if $x = x_0, \ldots, x_k = y$:

$p(x_{i-1}, x_i) > 0$, $i = 1, \ldots, k$

then $p_k(x, y)$

$$\geq \prod_{i=1}^{k} p(x_{i-1}, x_i) > 0.$$

Claim: ① if x, y: $x \to y$ but $y \not\to x$

then x is transient.

② if S is finite and x:

if $x \to y$ then $y \to x$ $\forall y \in S$

then x is recurrent.

① use $P_x(\cdots) = P(\cdots \mid X_0 = x)$,

then $P_x(T_x^+ = \infty) \geq P_x(X_n = y \text{ some } n)$

by assumption $x \to y$, $\exists\, n_0 :$

$$P_x(X_{n_0} = y) > 0$$

so

$$\rho_{xx} = 1 - P_x(T_x^+ = \infty)$$

$$\leq 1 - \underbrace{P_x(X_{n_0} = y)}_{>0} < 1.$$

$\therefore$ x transient.

reaches y, avoids x.

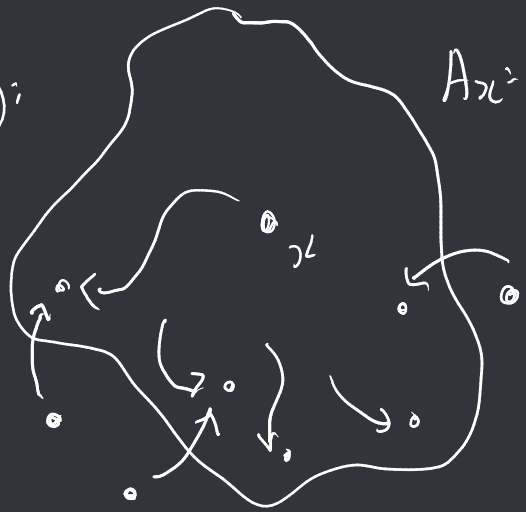
* missing step: must find event where

$$X_{n_0} = y \quad \text{some } n_0$$

$$\& \quad X_m \neq x \quad \forall \; 0 < m \leq n_0$$

(take shortest route to y).

(2):



$A_x := \{ y : x \to y \}$

by assumption,

$\forall y \in A_x,$

$\exists\, n_y : P_{n_y}(x,y) > 0.$

$P_y(T_x^+ \le n_y) > 0 \ldots$

finish next time